Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

An ideal battery of emf $2\ \mathrm{V}$ and a series resistance R are connected in the primary circuit of a potentiometer of length $1\ \mathrm{m}$ and resistance 5Ω. The value of R, to give a potential difference of $5\mathrm{mV}$ across $10\ \mathrm{cm}$ of potentiometer wire is

(A) 180Ω

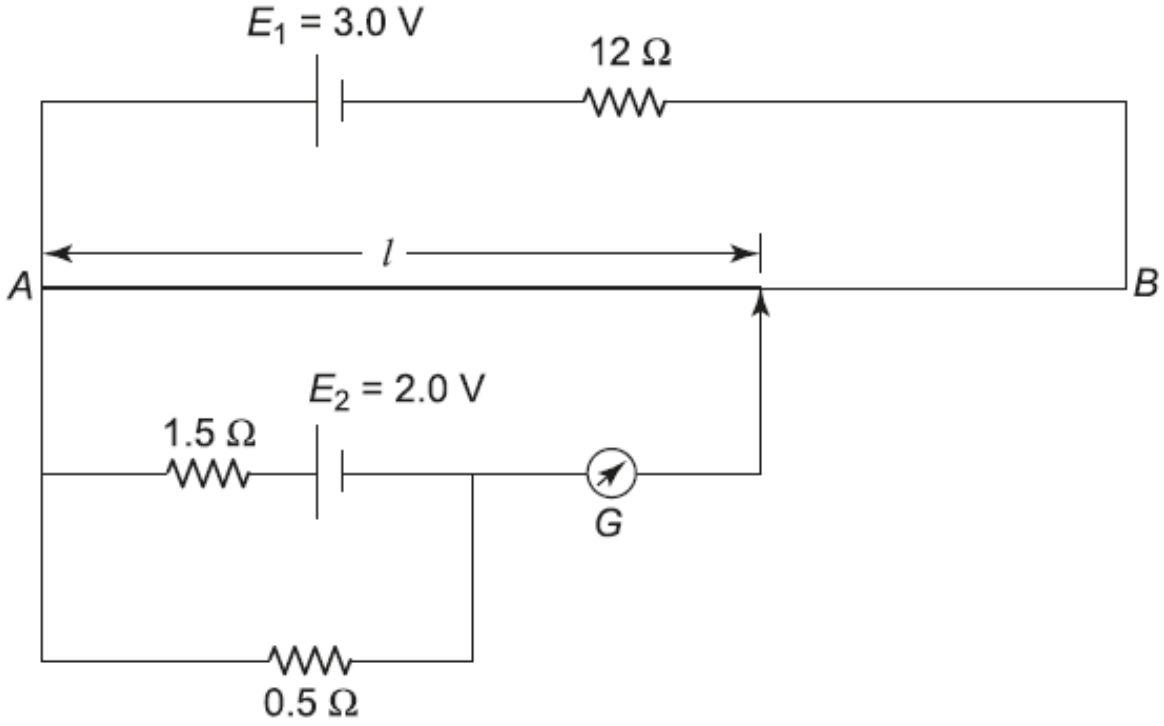
(B) 190Ω

(C) 195Ω

(D) 200Ω

Q.2

In the potentiometer circuit shown in Fig., find the value of l when the galvanometer shows no deflection. The length of wire AB is $100\ \mathrm{cm}$ and its resistance is 3Ω.



(A) 83.3 cm

(B) 53.3 cm

(C) 43.3 cm

(D) 183.3 cm

Q.3

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To measure the resistance of a galvanometer by half deflection method, a shunt is connected to the galvanometer. The shunt is

(A) low resistance connected in parallel

(B) low resistance connected in series

(C) high resistance connected in parallel

(D) high resistance connected in series

Q.4

A galvanometer of resistance 20Ω gives full scale deflection when a current of $1\ \mathrm{mA}$ is passed through it. What is the value of the shunt added to convert it into ammeter?

(A) 0.02 ohm

(B) 0.08 ohm

(C) 0.06 ohm

(D) 0.04 ohm

Q.5

Two resistors of 400Ω and 600Ω are connected in series with a 5.0 V battery of negligible internal resistance.
An ammeter of resistance 20Ω is used to measure the current in the circuit. Find the error in the measurement of current.

(A) 0.1 mA

(B) 0.8 mA

(C) 0.6 mA

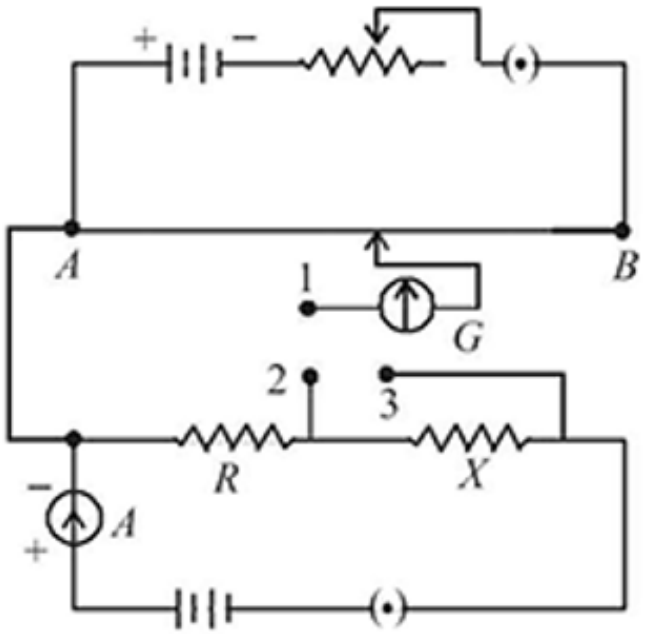
(D) 0.4 mA

Q.6

A potentiometer circuit is set up as shown. The potential gradient, across the potentiometer wire, is k volt $/\mathrm{cm}$ and the ammeter, present in the circuit, reads $1.0\ \mathrm{A}$ when two way key is switched off. The balance points, when the key between the terminals (i) 1 and 2 (ii) 1 and 3 , is plugged in, are found to be at lengths $1_1\ \mathrm{cm}$

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and l_2 cm respectively. The magnitudes, of the resistors R and X, in ohms, are then, equal, respectively, to



(A) $k(l_2 - l_1)$ and kl_2

(B) kl_1 and $k(l_2 - l_1)$

(C) $k(l_2 - l_1)$ and kl_1

(D) kl_1 and kl_2

Q.7

The length of a wire of a potentiometer is 100 cm, and the e.m.f. of its standard cell is E volt. It is employed to measure the e.m.f. of a battery whose internal resistance is 0.5Ω. If the balance point is obtained at $\ell =$ 30 cm from the positive end, the e.m.f. of the battery is

(A) $\frac{30E}{100.5}$

(B) $\frac{30E}{(100-0.5)}$

(C) $\frac{30(E-0.5i)}{100}$

(D) $\frac{30E}{100}$

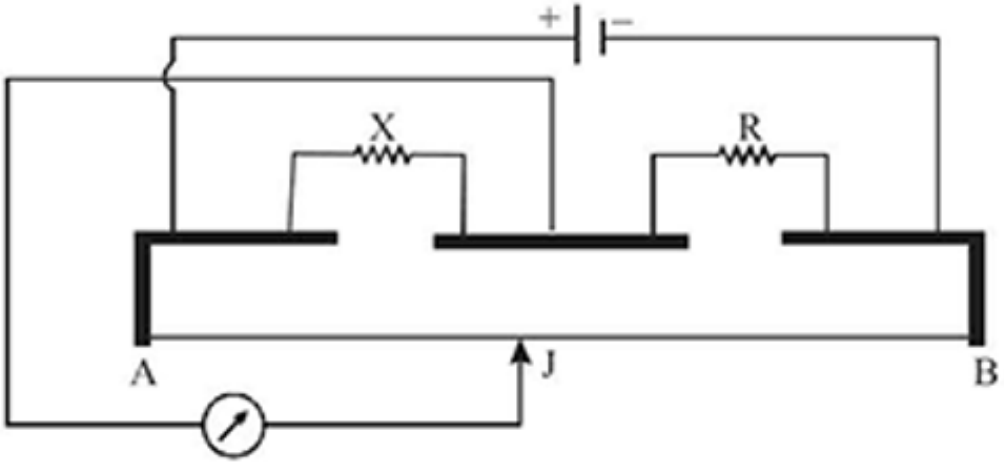
where i is the current in the potentiometer wire.

Q.8

The figure shows a meter-bridge circuit, $X = 12\Omega$ and $R = 18\Omega$. The jockey J is at the null point. If R is made 8Ω and the jockey J have to be moved by $4 \times A$cm to obtain null point again then find the value of

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A.



(A) 3

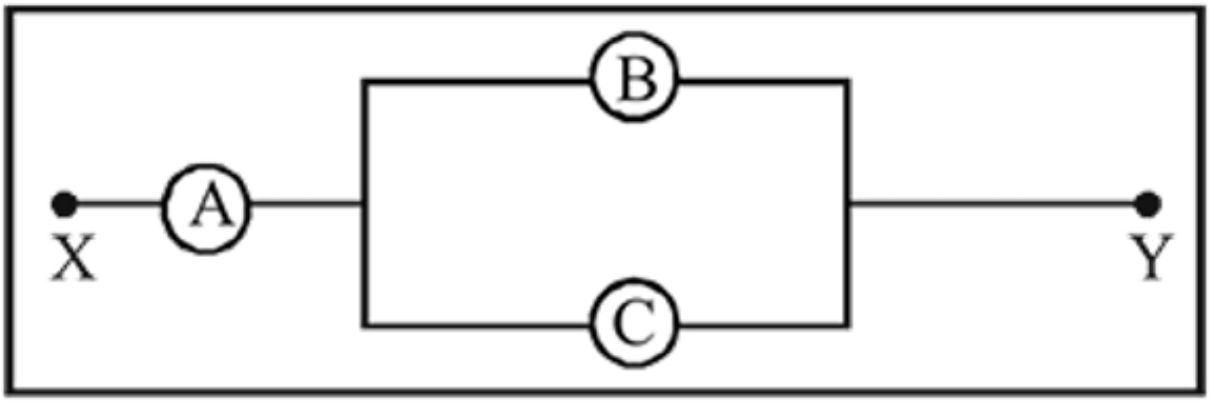
(B) 4

(C) 5

(D) 6

Q.9

A, B and C are voltmeters of resistance R, $1.5\mathrm{R}$ and $3\mathrm{R}$ respectively as shown in the figure. When some potential difference is applied between X and Y, the voltmeter readings are $\mathrm{V_A}$, $\mathrm{V_B}$ and $\mathrm{V_C}$ respectively. Then



(A) $\mathrm{V_A} \neq \mathrm{V_B} = \mathrm{V_C}$

(B) $\mathrm{V_A} \neq \mathrm{V_B} \neq \mathrm{V_C}$

(C) $V_A = V_B \neq V_C$

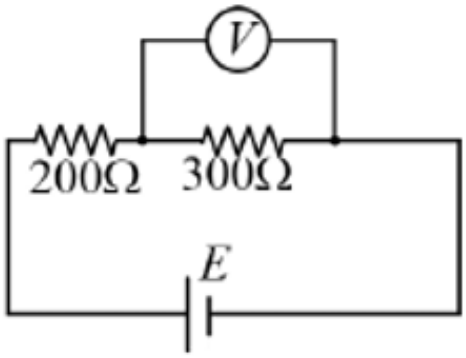
(D) $V_A = V_B = V_C$

Q.10

In the given circuit resistance of voltmeter is 400Ω

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and its reading is $20\ \text{V}$. Find the value of emf of battery



(A) $130/3$ volt

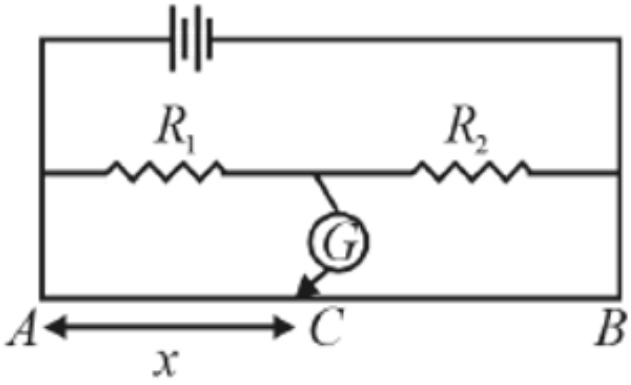
(B) 65 volt

(C) 40 volt

(D) 33.6 volt

Q.11

In the shown arrangement of the experiment of the meter bridge if AC corresponding to null deflection of galvanometer is x, what would be its value if the radius of the wire AB is doubled?



(A) x

(B) $x/4$

(C) $4x$

(D) (d) $2x$

Q.12

An ammeter is obtained by shunting a 30Ω galvanometer with a 30Ω resistance. What additional shunt should be connected across it to double the range?

(A) 15Ω

(B) 10Ω

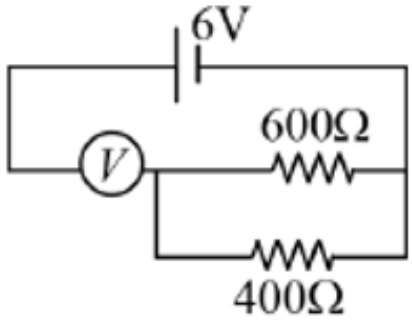
(C) 5Ω

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(D) none of these

Q.13

The measurement of voltmeter (ideal) in the following circuit is



(A) 2.4 V

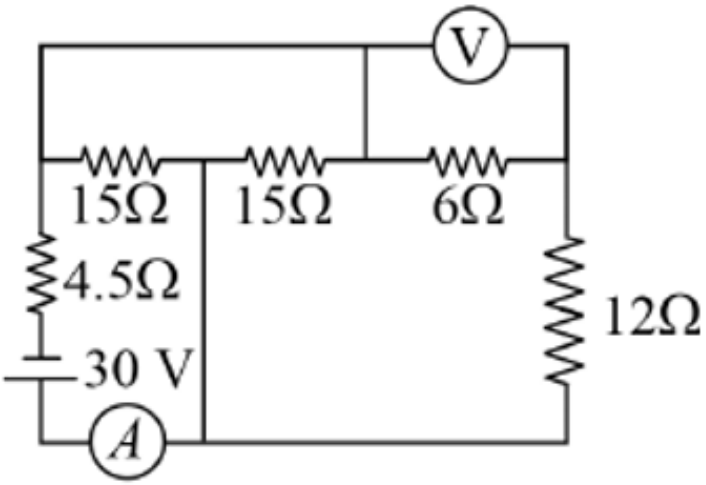
(B) zero

(C) 4.0 V

(D) 6.0 V

Q.14

A galvanometer of coil resistance 1Ω is converted into voltmeter by using a resistance of 5Ω in series and same galvanometer is converted into ammeter by using a shunt of 1Ω. Now ammeter and voltmeter connected in circuit as shown, find the reading of voltmeter and ammeter.



(A) 3 Volt, 3 amp

(B) 2 volt, 2amp

(C) 4 Volt, 3amp

(D) 3 volt, 4amp

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Q.15

An electric current passes through a circuit containing two wires of the same material connected in parallel. If the lengths of the wires are in the ratio $\frac{4}{3}$ and radius of the wires are in the ratio $\frac{2}{3}$, then the ratio of the current passing through the wires is $\frac{m}{n}$, where m and n are smallest integers. What is $m+n$?

(A) 4

(B) 5

(C) 6

(D) 7

Answers & Solutions

Q.1 Answer:

195Ω

Solution:

Current in the circuit is $I = \frac{E}{(R+r)} = \frac{2}{(R+5)}$.

Therefore, the potential difference across the potentiometer wire of length $L = 100$ cm **is**

$V = Ir = \frac{2}{(R+5)} \times 5 = \frac{10}{(R+5)}$ volt

$\therefore$ **Potential difference across** $l = 10$ cm **of the wire is**

$v = \frac{V \times l}{L} = \frac{10}{(R+5)} \times \frac{10}{100} = \frac{1}{(R+5)}$ volt

Given $v = 5\text{mV} = 5 \times 10^{-3}$ V. **Hence, we have**

$5 \times 10^{-3} = \frac{1}{(R+5)}$

which gives $R = 195\Omega$, **which is choice (c).**

Q.2 Answer:

83.3 cm

Solution:

Current flowing in AB **due to driver cell of emf** 3.0 V **is**

$I_1 = \frac{3.0}{(12+3)} = 0.2$ A

$\therefore$ **Potential difference across** AB **due to** E_1 **is**

$= 0.2 \times 3 = 0.6$ V

Potential difference across l **due to** E_1 **is** (l **is in** cm)

$V_l = \frac{0.6}{100} \times l$ **(1)**

Current through 0.5Ω **resistance due to** E_2 **is**

$I_2 = \frac{2.0}{(1.5+0.5)} = 1.0$ A

$\therefore$ **Potential difference across** 0.5Ω **resistance**

$= 0.5 \times 1.0 = 0.5$ V

At balance length l, **the potential difference across** l **due to** E_2 **is**

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$V_l' = 0.5$ V

At balance point $V_l = V_l'$. Equating (1) and (2), we have

$\frac{0.6}{100} \times l = 0.5 \Rightarrow l = 83.3$ cm

Q.3 Answer:

low resistance connected in parallel

Solution:

Shunt is a low resistance connected in parallel to galvanometer so that most of the current passes through the shunt, and also, the galvanometer is saved from the damages due to large currents.

Q.4 Answer:

0.02 ohm

Solution:

Given $I = 1.0$ A, $I_g = 1$ mA $= 00.1$ A and $G = 20\Omega$.

$S = \left(\frac{I_g}{I - I_g}\right) G = \left(\frac{0.001}{1.0 - 0.001}\right) \times 20 \simeq 0.02\Omega$

The required shunt resistance is 0.02Ω.

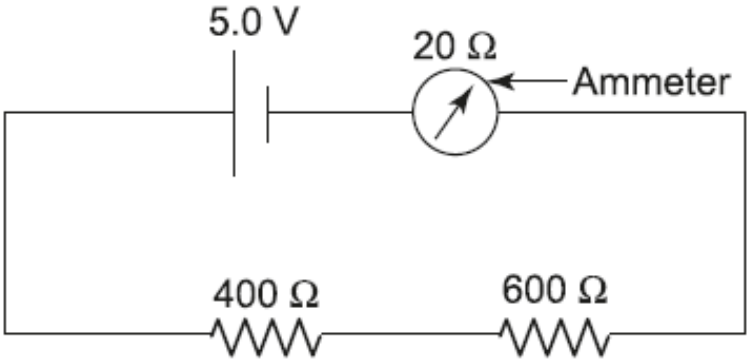
Q.5 Answer:

0.1 mA

Solution:

Actual current without ammeter in the circuit is (Fig.)

$I = \frac{5.0}{(400+600)} = 0.005$ A $= 5$ mA



When the ammeter is connected, the total resistance $= 400 + 600 + 20 = 1020\Omega$. The current indicated by the ammeter will be

$I' = \frac{5.0}{1020} = 0.0049$ A $= 4.9$ mA

Hence the ammeter reads less than the actual current. Error in the measurement of current $=$

5.0 mA $- 4.9$ mA $= 0.1$ mA

Q.6 Answer:

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$k l_1$ and $k\left(l_2 - l_1\right)$

Solution:

When key between the terminals 1 and 2 is plugged in,

P.D. across $R = IR = kl_1 \Rightarrow R = kl_1$ **as** $I = 1A$

(ii) When key between terminals 1 and 3 is plugged in, P.D. across$(X + R) = I(X + R) = kl_2$

$\Rightarrow X + R = kl_2$

$\therefore \quad X = k\left(l_2 - l_1\right)$

$\therefore \quad R = kl_1$ **and** $X = k\left(l_2 - l_1\right)$

Q.7 Answer:

$\frac{30E}{100}$

where i is the current in the potentiometer wire.

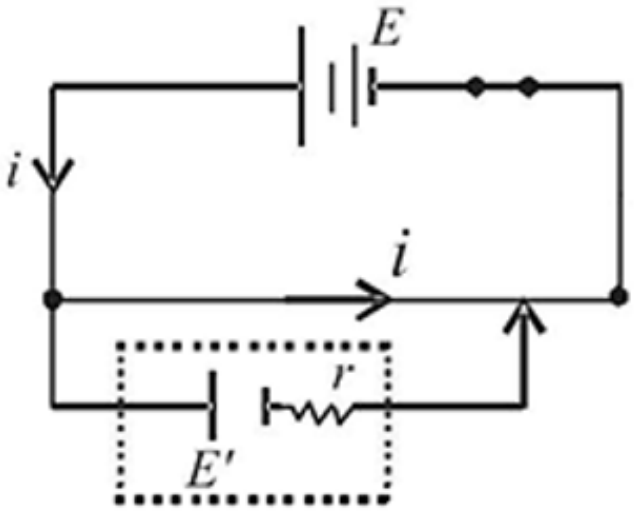
Solution:

From the principle of potentiometer, $\mathrm{V} \propto l$

$\Rightarrow \frac{\mathrm{V}}{\mathrm{E}} = \frac{l}{\mathrm{L}}$**; where**

$\mathrm{V} =$ **emf of battery,** $\mathrm{E} =$ **emf of standard cell.** $\mathrm{L} =$ **length of potentiometer wire**

$\mathrm{V} = \frac{\mathrm{E}l}{\mathrm{L}} = \frac{30\mathrm{E}}{100}$



Q.8 Answer:

5

Solution:

If $\ell_1 =$ **length from one end then** $\frac{\ell_1}{1-\ell_1} = \frac{X}{\mathrm{R}} = \frac{12}{18}$

$\ell_1 = \frac{12}{30}\ \mathrm{m} = 40\ \mathrm{cm}$

and $\ell_1' =$ **length from one end in second case** $\frac{\ell_1'}{1-\ell_1'} = \frac{X}{R'} = \frac{12}{8}$

$\ell_1' = \frac{12}{20}\ \mathrm{m} = 60\ \mathrm{cm}.$**; So shift** $= 20\ \mathrm{cm}$

Q.9 Answer:

$V_A = V_B = V_C$

Solution:

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As Resistance B and C are in parallel:

$$\mathrm{V_B} = \mathrm{V_C}$$

the equivalent resistance of resistor B and $\mathrm{C} = \frac{\mathrm{R_1 R_2}}{\mathrm{R_1 + R_2}}$

$$\mathrm{R_{eq}} = \left(\frac{1.5 \times 3}{1.5 + 3}\right) \times \mathrm{R} = \mathrm{R\Omega}$$

as resistance A and $\mathrm{R_{eq}}$ are equal, potential drop in each resistor will be equal. $\Rightarrow \mathrm{V}_A = \mathrm{V_B} = \mathrm{V_C}$

Q.10 Answer:

$130/3$ volt

Solution:

Current in voltmeter, $I = \frac{20}{400} = \frac{1}{20}$ amp
Current in $300\Omega = \frac{1}{15}\mathrm{amp}$
Current in $200\Omega = \frac{1}{20} + \frac{1}{15} = \frac{35}{300} = \frac{7}{60}$

$$E = 200 \times \frac{7}{60} + 20 = \frac{130}{3}\ \mathrm{V}$$

$\therefore$ (a)

Q.11 Answer:

x

Solution:

At null point,
$\frac{R_1}{R_2} = \frac{x}{(100-x)}$
If radius of the wire is doubled, then the resistance of AC will change and the resistance of CB will also change. But since $\frac{R_1}{R_2}$ does not change, so $\frac{x}{100-x}$ should also not change at null point.
Thus, null condition is independent of radius of AB wire. So when radius is doubled, null position will remain same i.e x

Q.12 Answer:

15Ω

Solution:

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Initially, $S = \frac{i_g}{I - i_g} G$

$30 = \frac{i_g}{I - i_g} \times 30 \Rightarrow I = 2i_g$

When range is doubled i.e $I' = 2I = 4i_g$

Now shunt required is: $S' = \frac{i_g}{I' - i_g} G = \frac{i_g}{4i_g - i_g} \times 30 = 10\Omega$

As shunt is connected in parallel to galvanometer, if additional shunt x **then** $30 \| x = 10$ **or**

$\frac{30x}{30+x} = 10$

$30 + x = 3x \Rightarrow x = 15\Omega$

Q.13 Answer:

6.0 V

Solution:

If the voltmeter is ideal then given circuit is an open circuit, so reading of voltmeter is equal to the e.m.f. of cell i.e., 6 V

$\therefore$

(d)

Q.14 Answer:

3 Volt, 3 amp

Solution:

$R_{\text{Voltmeter}} = 6\Omega, \quad R_{\text{ammeter}} = 0.5\Omega$

$R_{\text{eq}} = 10\Omega$

$I = \frac{30}{10} = 3A$

Reading of voltmeter $= 1 \times 3 = 3$ **volt**

- (a)

Q.15 Answer:

4

Solution:

Given that,

$\frac{l_1}{l_2} = \frac{4}{3}$ and $\frac{r_1}{r_2} = \frac{2}{3}$

Here, l_1 **and** l_2 **are the length of the wires while** r_1 **and** r_2 **are the radii of the wires.**

Now, we know that

$V = IR \Longrightarrow IR =$ constant

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$\Rightarrow I_1R_1 = I_2R_2$

or $\frac{I_1}{I_2} = \frac{R_2}{R_1} \quad \ldots .i$

But we know that the resistance of the wire is

$$R = \frac{\rho l}{A}$$

Hence, from Eq (i)

$$\frac{I_1}{I_2} = \frac{\rho l_2 / A_2}{\rho l_1 / A_1}$$

Here,

$R_1 = \frac{\rho l_1}{A_1}, R_2 = \frac{\rho l_2}{A_2} and\ \rho_1 = \rho_2 = \rho$

Because both wires are of same material.

$\therefore \frac{I_1}{I_2} = \frac{l_2 A_1}{l_1 A_2}$

$\Rightarrow \frac{I_1}{I_2} = \frac{l_2 \pi r_1^2}{l_1 \pi r_2^2}$

$\Rightarrow \frac{I_1}{I_2} = \frac{l_2 r_1^2}{l_1 r_2^2}$

Here, $\frac{l_2}{l_1} = \frac{3}{4}\ and \frac{r_1}{r_2} = \frac{2}{3}$

$\Rightarrow \frac{I_1}{I_2} = \frac{3}{4} \times \frac{2}{3}^2$

Or $\frac{I_1}{I_2} = \frac{1}{3} = \frac{m}{n} \Rightarrow m + n = 4$

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